

Mohammad Taha Askari

PHD CANDIDATE · ELECTRICAL AND COMPUTER ENGINEERING, THE UNIVERSITY OF BRITISH COLUMBIA

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Education

PhD in Electrical & Computer Engineering

Vancouver, Canada

ELECTRICAL AND COMPUTER ENGINEERING DEPARTMENT, THE UNIVERSITY OF BRITISH COLUMBIA (UBC)

Sep. 2022 - Apr. 2027

- Advisor: Prof. Lutz Lampe

M.A.Sc. in Electrical & Computer Engineering

Vancouver, Canada

ELECTRICAL AND COMPUTER ENGINEERING DEPARTMENT, THE UNIVERSITY OF BRITISH COLUMBIA (UBC)

Sep. 2020 - Aug. 2022

- Cumulative GPA: 93.7 out of 100 (4.0/4.0) via 18 credits
- Advisor: Prof. Lutz Lampe

B.Sc. in Electrical Engineering

Tehran, Iran

DEPARTMENT OF ELECTRICAL ENGINEERING, SHARIF UNIVERSITY OF TECHNOLOGY (SUT)

Sep. 2015 - Jul. 2020

- Cumulative GPA: 18.42 out of 20 (4.0/4.0) via 146 credits
- Advisor: Prof. Hamid Behroozi

Research Interests

Machine Learning for Sequential and Structured Data

Probabilistic Modeling and Joint Distribution Learning

Neural Sequence Models (Autoregressive Models, Transformers)

End-to-End Learning and Optimization

Information Theory and Statistical Inference

Research Experience

M.A.Sc. & PhD Thesis at Data Communications Group

ECE Department, UBC

RESEARCH ASSISTANT (SUPERVISOR: PROF. LUTZ LAMPE)

Sep. 2020 - Apr. 2027

- Conducted independent research on **probabilistic modeling and sequence design** for communication systems with memory and nonlinear dynamics.
- Developed an **analytical low-pass filter model** to characterize interactions between probabilistic shaping and nonlinear optical fiber channels, enabling interpretable system-level optimization.
- Proposed a **sequence selection algorithm** that improves robustness to nonlinear interference by exploiting temporal dependencies.
- Designed a **Bayesian carrier phase recovery algorithm** incorporating prior distributions, achieving improved performance in low-SNR regime.
- Investigated **end-to-end learning approaches** using sequence-based autoencoder architectures for probabilistic shaping and joint distribution learning.

Research Internship at Nokia Bell Labs

Greater Paris Metropolitan Region, France

MACHINE LEARNING RESEARCH INTERN

April. 2025 - April. 2026

- Designed and implemented **Neural Probabilistic Shaping (NPS)**, an autoregressive **sequence modeling framework** using neural sequence models and **Gumbel-Softmax** sampling for joint distribution learning.
- Built an **end-to-end differentiable training pipeline** integrating neural models with differentiable channel models, enabling gradient-based optimization over sequences.
- Evaluated the proposed models on **multi-dimensional settings**, demonstrating improved robustness and scalability compared to heuristic and sampling-based methods.

Research Internship at Roche Canada

Greater Toronto Area, Canada

ALGORITHM R&D SOFTWARE ENGINEERING INTERN

Jun. 2024 - Oct. 2024

- Designed and implemented **ML pipelines** for feature extraction, dataset generation, and neural network training for large-scale sequencing data.
- Developed and optimized **weakly supervised deep learning models**, including a custom **multi-label architecture** for multi-class anomaly detection and segmentation.
- Integrated **model interpretability, performance analysis, and computational complexity evaluation** into the training workflow to support iterative model improvement.

Publications and Posters

- M.T. Askari, L. Lampe, and A. Ghazisaeidi "Neural Probabilistic Shaping: Joint Distribution Learning for Optical Fiber Communications," 2025 European Conference on Optical Communication (ECOC). [Online Access]
- M.T. Askari and L. Lampe, "Probabilistic Shaping for Nonlinearity Tolerance," Journal of Lightwave Technology. [Online Access]
- M.T. Askari and L. Lampe, "Perturbation-based Sequence Selection for Probabilistic Amplitude Shaping," 2024 European Conference on Optical Communication (ECOC). [Online Access]
- M.T. Askari and L. Lampe, "Bayesian Phase Search for Probabilistic Amplitude Shaping," 2023 European Conference on Optical Communication (ECOC). [Online Access]
- M.T. Askari, "Interplay between Fiber Nonlinearity and Probabilistic Amplitude Shaping," Master's Thesis, UBC [Online Access]
- M.T. Askari, L. Lampe, and J. Mitra, "Probabilistic Amplitude Shaping and Nonlinearity Tolerance: Analysis and Sequence Selection Method," Journal of Lightwave Technology. [Online Access]
- M.T. Askari, L. Lampe, and J. Mitra, "Nonlinearity Tolerant Shaping with Sequence Selection," 2022 European Conference on Optical Communication (ECOC) [Online Access]
- M.T. Askari, "Nonlinearity Tolerant Sequence Selection," Poster at 17th Canadian Workshop on Information Theory (CWIT)

Language Skills

English Advanced
Persian Native

Relevant Courses

CPSC 436N-Natural Language Processing	95.0 out of 100	Dr. Shwartz	UBC	Winter1 2023
EECE 571D-Detection, Estimation, and Learning	98.0 out of 100	Dr. Lampe	UBC	Winter2 2021
EECE 562-Statistical Signal Processing	90.0 out of 100	Dr. Wang	UBC	Winter1 2021
CPSC 540-Advanced Machine Learning	96.0 out of 100	Dr. Schmidt	UBC	Winter2 2020
CPSC 340-Machine Learning & Data Mining	100 out of 100	Dr. Wood	UBC	Winter1 2020
EECE 565-Communication Networks	95.0 out of 100	Dr. Wong	UBC	Winter1 2020
Introduction to Machine Learning	18.0 out of 20	Dr. Mohammadzade	SUT	Spring 2019
Information & Coding Theory	18.7 out of 20	Dr. Mirmohseni	SUT	Fall 2018

Software Proficiency

Programming Python, C/C++, Git, SQL, Matlab, Julia, R

Teaching Experience

Teaching Assistant	STAT 251-Elementary Statistics	Dr. Premarathna	UBC	Winter2 2024
Teaching Assistant	STAT 200-Elementary Statistics for Applications	Dr. Lourenzutti	UBC	Winter1 2023
Teaching Assistant	ELEC 221-Signals & Systems	Dr. Thrampoulidis	UBC	Winter1 2023
Teaching Assistant	STAT 200-Elementary Statistics for Applications	Dr. Burr	UBC	Summer1 2023
Teaching Assistant	STAT 251-Elementary Statistics	Dr. Premarathna	UBC	Winter2 2022
Teaching Assistant	ELEC 221-Signals & Systems	Dr. Di Matteo	UBC	Winter1 2022
Teaching Assistant	STAT 200-Elementary Statistics for Applications	Dr. Burr	UBC	Winter1 2022
Teaching Assistant	CPSC 330-Applied Machine Learning	Dr. Oveisi	UBC	Summer1 2022
Teaching Assistant	STAT 306-Finding Relationships in Data	Dr. Premarathna	UBC	Winter1 2021
Teaching Assistant	CPSC 303-Numerical Approximation	Dr. Yi	UBC	Winter2 2020

Voluntary Experience

ECE Graduate Student Association (ECEGSA)

ECE Department, UBC

VP ACADEMIC

Apr. 2024-Present

- Organized academic and professional development events, including resume workshops, career fairs, and lightning talks, to support students' career advancement and academic success.
- Facilitated opportunities for graduate students to build their professional brand and network with industry professionals and academic peers.

Online Instructional Skills Workshop

CIRTL, UBC

PARTICIPANT

Jan. 2021

- Practiced developing online lesson plans and using techniques to involve students actively in the online learning process.
- Taught three lessons to employ new teaching techniques.

Conference on "Emerging Technologies for Electric Vehicles and Smart Cars"

EE Department, SUT

ADMINISTRATIVE HEAD

Nov. 2018

- Directed a team of ten students in holding the conference.
- Gained knowledge and experience in several managerial skills like apportioning responsibilities and group management.

International Biology Olympiad

Tehran, Iran

TEAM GUIDE

Jul. 2018

- Guided Czech Republic's delegation in the 29th International Biology Olympiad (IBO).
- Acquired experience in the communication skills required for accompanying the delegation.
- Awarded certification from the IBO committee.

Honors & Awards

Awarded UBC 2025 PhD Graduate Student Initiative (GSI) with an amount of \$8,200.

2025

Nominated for UBC Four Year Doctoral Fellowship with an amount of \$18,200 per year.

2023

Awarded UBC 2021 M.A.Sc. Graduate Student Initiative (GSI) with an amount of \$5,000.

2021

Member of the National Elites Foundation by being in the best ranks of the University entrance exam.

2015-Present

Selected Course Projects

COVID-19 Detection: A Combination of Transformer Models with Image Augmentation Methods

Winter 2020

COURSE: ADVANCED MACHINE LEARNING

UBC

- Studied methods for generating synthetic X-ray images to improve the accuracy of COVID-19 detection models and implemented GAN and VAE to augment data.
- Applied transformer models to COVID-19 classification based on chest X-ray images.
- Supervisor: Dr. Schmidt

Trajectory Prediction

Winter 2020

COURSE: MACHINE LEARNING & DATA MINING

UBC

- Studied methods for autonomous driving trajectory prediction and implemented a chained neural network model.
- Supervisor: Dr. Wood